# REVISION TABLEAU DE KARNAUGH

## Exercice 1

$$F_1 = \bar{a} . b. \bar{c} + \bar{a} . b. c$$

- Simplification par la méthode algébrique

$F_1 = \bar{a} . b. \bar{c} + \bar{a} . b. c$
$F_1 = \bar{a} . b. ( \bar{c} + c )$
$F_1 = \bar{a} . b. 1$
$F_1 = \bar{a} . b$

- Simplification par les tableaux de Karnaugh

| $F_1$ | a b | | | |
|---|---|---|---|---|
| | 00 | 01 | 11 | 10 |
| c = 0 | 0 | 1 | 0 | 0 |
| c = 1 | 0 | 1 | 0 | 0 |

$F_1 = \bar{a} . b$

## Exercice 2

$$F_2 = \bar{a} . \bar{b}. \bar{c} + \bar{a} . \bar{b} . c + a .b. \bar{c} + a . b. c$$

- Simplification par la méthode algébrique

$F_2 = \bar{a} . \bar{b}. \bar{c} + \bar{a} . \bar{b} . c + a .b. \bar{c} + a . b. c$
$F_2 = \bar{a} . \bar{b}. ( \bar{c} + c ) + a .b. ( \bar{c} + c)$
$F_2 = \bar{a} . \bar{b}. 1 + a .b. 1$
$F_2 = \bar{a} . \bar{b} + a . b$

- Simplification par les tableaux de Karnaugh

| $F_2$ | a b | | | |
|---|---|---|---|---|
| | 00 | 01 | 11 | 10 |
| c = 0 | 1 | 0 | 1 | 0 |
| c = 1 | 1 | 0 | 1 | 0 |

$F_2 = \bar{a} . \bar{b} + a . b$

http://electrocorot.free.fr/

## Exercice 3

$$F_3 = a \cdot b \cdot \bar{c} + a \cdot b \cdot c + a \cdot \bar{b} \cdot \bar{c} + a \cdot \bar{b} \cdot c$$

➢ Simplification par la méthode algébrique

$F_3 = a \cdot b \cdot \bar{c} + a \cdot b \cdot c + a \cdot \bar{b} \cdot \bar{c} + a \cdot \bar{b} \cdot c$

$F_3 = a \cdot b \cdot (\bar{c} + c) + a \cdot \bar{b} \cdot (\bar{c} + c)$

$F_3 = a \cdot b \cdot 1 + a \cdot \bar{b} \cdot 1 = a \cdot b + a \cdot \bar{b}$

$F_3 = a (b + \bar{b})$

$F_3 = a \cdot 1$

$F_3 = a$

➢ Simplification par les tableaux de Karnaugh

| $F_3$ | a b | | | |
|---|---|---|---|---|
| c | 00 | 01 | 11 | 10 |
| 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 | 1 |

$F_3 = a$

## Exercice 4

$$F_4 = a \cdot b \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot c + a \cdot \bar{b} \cdot \bar{c} + a \cdot \bar{b} \cdot c$$

➢ Simplification par la méthode algébrique

$F_4 = a \cdot b \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot c + a \cdot \bar{b} \cdot \bar{c} + a \cdot \bar{b} \cdot c$

$F_4 = a \cdot \bar{c} (b + \bar{b}) + \bar{b} \cdot c (\bar{a} + a)$

$F_4 = a \cdot \bar{c} \cdot 1 + \bar{b} \cdot c \cdot 1$

$F_4 = a \cdot \bar{c} + \bar{b} \cdot c$

➢ Simplification par les tableaux de Karnaugh

$F_4 = a \cdot b \cdot \bar{c} + \bar{a} \cdot \bar{b} \cdot c + a \cdot \bar{b} \cdot \bar{c} + a \cdot \bar{b} \cdot c$

| $F_4$ | a b | | | |
|---|---|---|---|---|
| c | 00 | 01 | 11 | 10 |
| 0 | 0 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 |

$F_4 = a \cdot \bar{c} + \bar{b} \cdot c$

http://electrocorot.free.fr/

## Exercice 5

$$F_5 = a \cdot b \cdot c + \overline{a} \cdot b \cdot c + \overline{a} \cdot \overline{b} \cdot c + a \cdot b \cdot \overline{c}$$

➢ Simplification par la méthode algébrique

$F_5 = a \cdot b \cdot c + \overline{a} \cdot b \cdot c + \overline{a} \cdot \overline{b} \cdot c + a \cdot b \cdot \overline{c}$

$F_5 = a \cdot b \cdot (c + \overline{c}) + \overline{a} \cdot c \ (b + \overline{b})$

$F_5 = a \cdot b \cdot 1 + \overline{a} \cdot c \cdot 1$

$F_5 = a \cdot b + \overline{a} \cdot c$

➢ Simplification par les tableaux de Karnaugh

| $F_5$ | | a b | | | |
|---|---|---|---|---|---|
| | | 00 | 01 | 11 | 10 |
| c | 0 | 0 | 0 | 1 | 0 |
| | 1 | 1 | 1 | 1 | 0 |

$F_5 = a \cdot b + \overline{a} \cdot c$

## Exercice 6

$$F_6 = \overline{a} \cdot \overline{b} \cdot c \cdot \overline{d} + \overline{a} \cdot b \cdot \overline{c} \cdot d + a \cdot \overline{b} \cdot c \cdot \overline{d} + \overline{a} \cdot \overline{b} \cdot \overline{c} \cdot \overline{d} + a \cdot b \cdot \overline{c} \cdot d + \overline{a} \cdot b \cdot c \cdot d + a \cdot \overline{b} \cdot \overline{c} \cdot \overline{d}$$

➢ Simplification par la méthode algébrique

$F_6 = \overline{a} \cdot \overline{b} \cdot c \cdot \overline{d} + \overline{a} \cdot b \cdot \overline{c} \cdot d + a \cdot \overline{b} \cdot c \cdot \overline{d} + \overline{a} \cdot \overline{b} \cdot \overline{c} \cdot \overline{d} + a \cdot b \cdot \overline{c} \cdot d + \overline{a} \cdot b \cdot c \cdot d + a \cdot \overline{b} \cdot \overline{c} \cdot \overline{d}$

$F_6 = a \cdot b \cdot \overline{c} \cdot d +$

$F_6 = \overline{a} \cdot \overline{b} \cdot \overline{d} \ (c + \overline{c}) + \overline{a} \cdot b \cdot d \cdot (\overline{c} + c) + a \cdot \overline{b} \cdot \overline{d} \ (c + \overline{c}) + a \cdot b \cdot \overline{c} \cdot d$

$F_6 = \overline{a} \cdot \overline{b} \cdot \overline{d} \ .1 + \overline{a} \cdot b \cdot d \cdot 1 + a \cdot \overline{b} \cdot \overline{d} \ .1 + a \cdot b \cdot \overline{c} \cdot d$

$F_6 = \overline{a} \cdot \overline{b} \cdot \overline{d} + \overline{a} \cdot b \cdot d + a \cdot \overline{b} \cdot \overline{d} + a \cdot b \cdot \overline{c} \cdot d$

$F_6 = \overline{b} \cdot \overline{d} \ (\overline{a} + a.) + b \cdot d \ (\overline{a} + a \cdot \overline{c})$

$F_6 = \overline{b} \cdot \overline{d} \ .1 + b \cdot d \ (\overline{a} + a \cdot \overline{c})$

$F_6 = \overline{b} \cdot \overline{d} + b \cdot d \ (\overline{a} + a \cdot \overline{c})$

$F_6 = \overline{b} \cdot \overline{d} + b \cdot d \ (\overline{a} + \overline{c})$

➢ Simplification par les tableaux de Karnaugh

| $F_6$ | | a b | | | |
|---|---|---|---|---|---|
| | | 00 | 01 | 11 | 10 |
| | 00 | 1 | 0 | 0 | 1 |
| c | 01 | 0 | 1 | 1 | 0 |
| d | 11 | 0 | 1 | 0 | 0 |
| | 10 | 1 | 0 | 0 | 1 |

$F_6 = \overline{b} \cdot \overline{d} + b \cdot \overline{c} \cdot d + \overline{a} \cdot b \cdot d$

$= \overline{b} \cdot \overline{d} + b \cdot d \ (\overline{a} + \overline{c})$

http://electrocorot.free.fr/

## Exercice 7

$$F_7 = a \cdot \overline{b} \cdot c + \overline{d} + c \cdot d$$

➢ Simplification par les tableaux de Karnaugh

***F7***

| c \ a b | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 1 | 1 | 1 |
| 01 | 0 | 0 | 0 | 0 |
| 11 | 1 | 1 | 1 | 1 |
| 10 | 1 | 1 | 1 | 1 |

$$F_7 = \overline{c} \cdot \overline{d} + c$$

## Exercice 8

$$F_8 = \overline{a} \cdot b + c \cdot \overline{d}$$

➢ Simplification par les tableaux de Karnaugh

***F8***

| c \ a b | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 0 | 0 |
| 01 | 0 | 1 | 0 | 0 |
| 11 | 0 | 1 | 0 | 0 |
| 10 | 1 | 1 | 1 | 1 |

$$F_8 = \overline{a} \cdot b + c \cdot \overline{d}$$

http://electrocarot.free.fr/